



Semester I Examinations 2012/ 2013

Exam Code(s) 1MF, 1SD, 1MDM
Exam(s) Masters in Software Design & Development
Higher Diploma in Applied Science (Software Design & Development)
Masters in Digital Media

Module Code(s) CT870
Module(s) Internet Programming

Paper No. I
Repeat Paper

External Examiner(s) Prof. M. O'Boyle
Internal Examiner(s) Prof. G. Lyons
Dr. M. Madden
* Dr. S. Redfern

Instructions: Answer any three questions.
All questions carry equal marks.

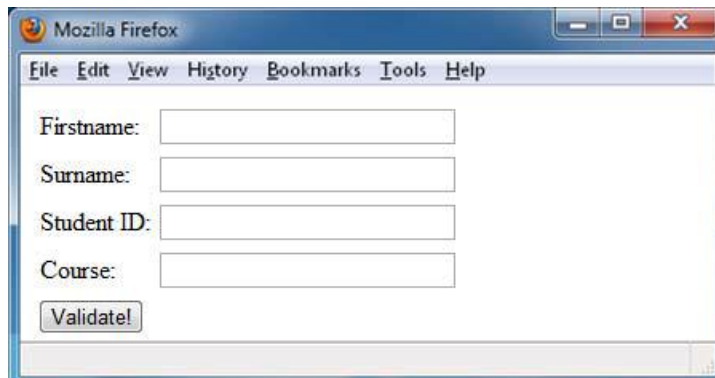
Duration 2 hours
No. of Pages 6
Department(s) Information Technology
Course Co-ordinator(s)

Requirements:

MCQ Release to Library: Yes ☐ No ☐
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Q1.

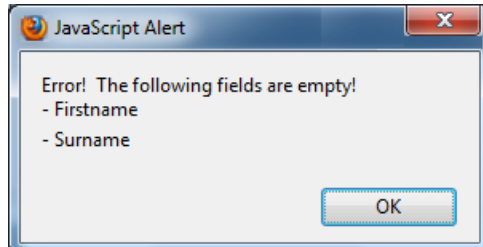
- a) With regard to HTML forms, what is the purpose of data validation? What are the advantages and disadvantages of server side and client side validation? Why might it be sensible to validate data twice, i.e. on both client and server? [6 marks]
- b) Write HTML code to produce the form as shown below.



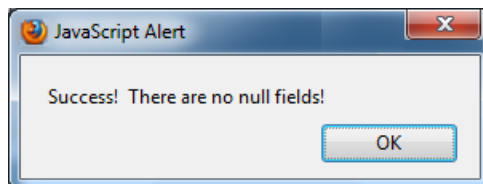
The screenshot shows a Mozilla Firefox browser window. The address bar is empty. The menu bar includes File, Edit, View, History, Bookmarks, Tools, and Help. The form contains four text input fields labeled 'Firstname:', 'Surname:', 'Student ID:', and 'Course:'. Below the fields is a button labeled 'Validate!'.

[3 marks]

When the Validate button is clicked, use client side validation to ensure that no fields are left null. If any fields are left null, an alert box similar to below should appear, informing the user which fields were left empty. (NB: It is not necessary to validate for data type, merely for null (blank) strings).



If all fields are completed, an alert box similar to the one shown below should appear



[5 marks]

- c) Write a simple webpage which uses Javascript to present a clock which updates once per second. The clock should be presented in this form:

The time is: **12:06:23**

[6 marks]

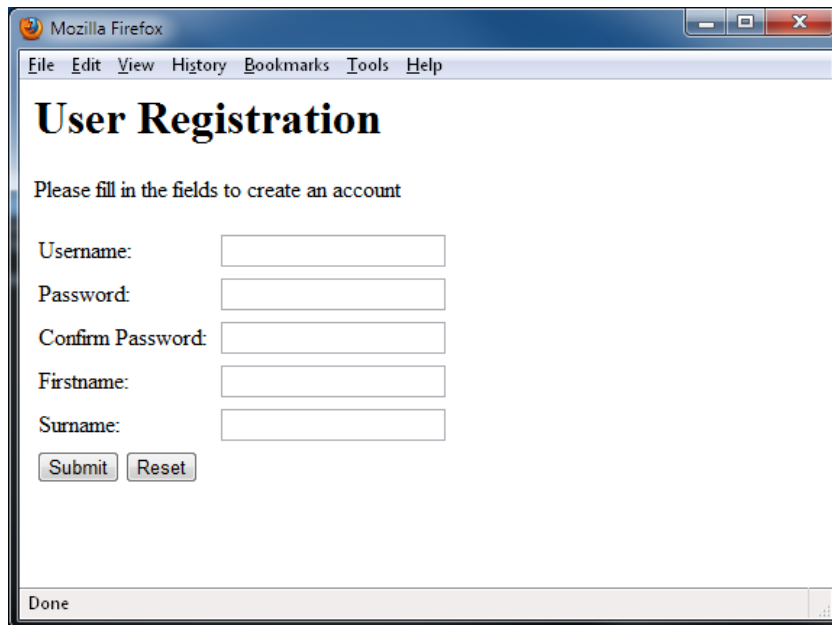
Q2.

- a) Compare and contrast the following ways of deploying CSS to apply style to HTML elements. Use code samples to illustrate how you would apply each of these: (i) Inline Style, (ii) Embedded Style, (iii) External Style Sheet [5 marks]
- b) Create an HTML user registration form. The form will accept a user's registration details, and be posted to a PHP page which will insert the data into a MySQL database. Assume a MySQL database exists on the webserver, with a table called 'users' containing the following fields:

Fieldname	Datatype	Description
id	int(10)	A unique key for each user
username	varchar(50)	The user's username
password	varchar(50)	The user's plaintext password
firstname	varchar(50)	The user's first name
surname	varchar(50)	The user's surname

NB: Assume the password is stored as plaintext.

Write HTML code to produce a user registration form similar to the one below to accept a username and password from a user. [3 marks]



Ensure that the registration form is validated server side so that the following conditions are met:

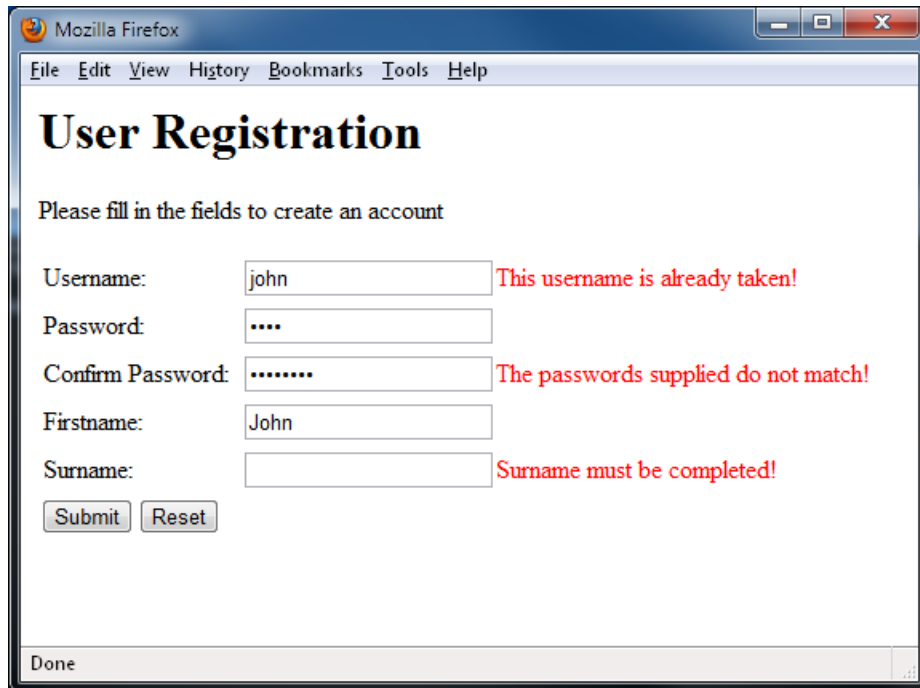
- No fields are null
 - The desired Username is not already taken
 - The Password and Confirm Password fields contain the same text
- [5 marks]

Q2. is continued on the next page

Q2. continued from the previous page

In the event of errors, messages should be displayed to the user highlighting the problematic field(s). Valid entries should be automatically filled into the text boxes so that they do not have to re-enter any fields that passed validation.

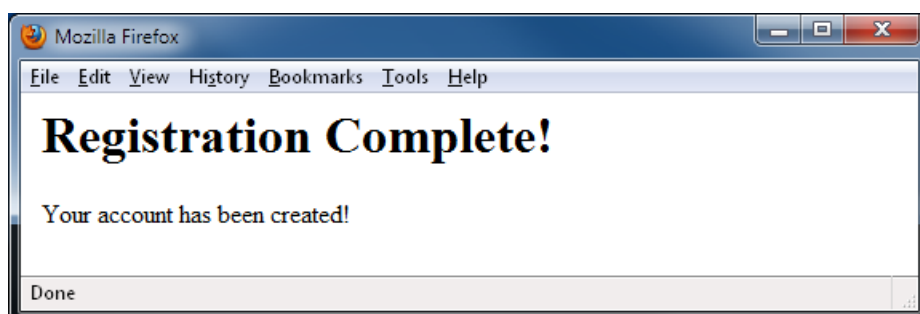
[4 marks]



The screenshot shows a Mozilla Firefox browser window with the title 'User Registration'. The page content includes a heading 'User Registration' and a subheading 'Please fill in the fields to create an account'. Below this, there are five input fields: 'Username:' with the value 'john', 'Password:' with four dots, 'Confirm Password:' with seven dots, 'Firstname:' with the value 'John', and 'Surname:' which is empty. To the right of the 'Username:' field, there is a red error message: 'This username is already taken!'. To the right of the 'Confirm Password:' field, there is a red error message: 'The passwords supplied do not match!'. To the right of the 'Surname:' field, there is a red error message: 'Surname must be completed!'. At the bottom of the form, there are two buttons: 'Submit' and 'Reset'. The status bar at the bottom of the browser window shows 'Done'.

On successful validation of the form, the PHP script should insert the new details into the users table in the database. The user should also be presented with a message to confirm that the registration has completed successfully

[3 marks]



The screenshot shows a Mozilla Firefox browser window with the title 'Registration Complete!'. The page content includes a heading 'Registration Complete!' and a subheading 'Your account has been created!'. The status bar at the bottom of the browser window shows 'Done'.

Q3.

- a) PHP has an inbuilt function for sending emails, as shown below. Identify and explain each of the four parameters sent to this function. Write PHP code which uses this function to send a short email message.

```
mail($to, $subject, $body, $headers);
```

[5 marks]

- b) Assuming an online movie shop uses a MySQL database which contains a table called 'movies' with the following schema:

<u>Fieldname</u>	<u>Datatype</u>	<u>Description</u>
id	int(10)	A unique key for each movie
title	varchar(50)	The title of the movie
price	float	The price of the movie (in Euro)
quantity	int(10)	The stock level, i.e. quantity in the shop
image	varchar(50)	The filename of an image of the movie's artwork

- Write PHP code to connect to the database and select all movies which cost less than 10 Euro. [3 marks]
- Extend this code so that the resulting dataset is displayed in an HTML page, with the details neatly laid out using a table. You can assume that the artwork images exist on the server, in a subfolder called 'images' [5 marks]
- Further extend this code so that the price used in the query, rather than being fixed at 10 Euro, is instead received by the PHP page either by a POSTed form or a querystring. You do not need to construct the HTML page which posts this data.

[3 marks]

- c) Briefly explain each of the following methods of the Window object in the HTML Document Object Model (DOM):

- `window.alert( msg );`
- `window.setTimeout( code, delay );`
- `window.confirm( msg );`

[4 marks]

Q4.

- a) How can Javascript and DHTML be used to perform a simple animation which moves an image across the screen? Illustrate your answer with sample code. [4 marks]

- b) Consider the “7 Times Table” pictured below.

Times Tables!

7 Times Table	
1 x 7 =	7
2 x 7 =	14
3 x 7 =	21
4 x 7 =	28
5 x 7 =	35
6 x 7 =	42
7 x 7 =	49
8 x 7 =	56
9 x 7 =	63
10 x 7 =	70

- Write the HTML code necessary to display this table on a webpage, paying attention to styling of the table and text [3 marks]

- Explain in general terms, how might Javascript be used to automatically produce such a “times table” on a webpage when the user has typed a number into the text-box? What role might a <DIV> tag have in this process? [2 marks]

- Write suitable Javascript or PHP code to achieve this automatic “times table” production, noting any assumption made about the HTML form which contains the text box. [5 marks]

- c) Assuming that you have been supplied the three .gif images shown below:

- Write suitable HTML/CSS code to display the “one.gif” image in the centre of the screen. [2 marks]
- Write suitable Javascript code to achieve the following: (i) when the image is clicked it should be replaced with “two.gif”. (ii) When the image is clicked again it should be replaced with “three.gif”. (iii) On the third click, “one.gif” should be used again. Note that all changes should be done on the client using JavaScript, i.e. a new HTML page should not be requested from the server with each click. [4 marks]

